RITIKA GAUR

Bulandshahr, Uttar Pradesh

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EDUCATION

G.L. Bajaj Institute of Technology and Management **2023 – 2027**

B.Tech - Computer Science & Engineering (CGPA: 8.4) *Greater Noida, Uttar Pradesh*

Azad Public School **2021 – 2022**

Senior Secondary (Percentage: 93.2) *Bulandshahr, Uttar Pradesh*

Saint Momina School **2019-2020**

High School (Percentage: 91.2) *Bulandshahr, Uttar Pradesh*

RELEVANT COURSEWORK

• Data Structures & Algorithms • Database Management Systems (DBMS)	• Object-Oriented System Design (OOSD) • Object-Oriented Programming (OOPs)	• Operating Systems

PROJECTS

Pollination Activity Detector

- Developed an AI-based poillination prediction system using XGBoost to classify pollination levels based on real-time weather data.
- Integrated OpenWeatherMap API for live environmental data and Flask REST API for backend processing with real-time insights to assists farmers.
- Built an interactive frontend using HTML, CSS and JavaScript with voice feedback using Speech Synthesis API.

HealOPS

- Designed a Self-Healing Application System that monitors application health using Prometheus, detects anomalies using Python-based ML logic (Isolation Forest) and automatically triggers recovery actions.
- Built a Dockerized Flask application and used docker-compose to manage services, enabling automatic restarts of the application when abnormal behaviour (CPU spikes, request overload) is detected.
- Implemented system observability using Grafana for visualization and a fault injection module to simulate failures, validating automated recovery workflows in a controlled environment

TECHNICAL SKILLS

Languages: C++, Python.
Backend: Flask, REST APIs.
Machine Leaning: XGBoost, Random Forest, Isolation Forest.
Frontend: HTML, Tailwind CSS, JavaScript.
Databases: MySQL.
DevOps: Docker, Docker Compose, Prometheus, Grafana.
Programming Concepts: OOSD Concepts, OOPs Concepts.

EXTRACURRICULAR ACTIVITIES AND SOFT SKILLS:

- Member (Abhyudaya Club): Led cultural and technical events, managed event planning.
- Time Management, Leadership, Communication, Teamwork.